

Name: _____ Student ID: _____

HOMEWORK 4

Continuous-time Fourier Transform (Part 2)

2110203 COMPUTER ENGINEERING MATHEMATICS II

Generative AI Usage Declaration

- ☐ No AI Used: I did not use any AI tools to complete this homework.
- ☐ AI Assisted: I used AI tools as detailed below:

Where	AI model	How

Student Acknowledgement : I declare that I have reviewed, understood, and tested all AI-generated output. I take full responsibility for the accuracy, logic, and functionality of the final submitted work.

Signature: _____ Date: _____

Name: _____

Student ID: _____

Write solutions and give explanations (1 point)

SEND ONLY PROBLEM 1. and 2.

1. (0.5 points, 0.1 points each) Let:

$$\mathcal{F}\{x(t)\} = X(j\omega) = \text{rect}\left(\frac{\omega - 1}{2}\right)$$

Find Fourier transform of

1.1. $x(-2t + 4)$

1.2. $(t - 1)x(t - 1)$

1.3. $t \frac{dx(t)}{dt}$

Name: _____

Student ID: _____

1.4. $x(2t - 1) \exp[-j2t]$

1.5. $x(t) * x(t - 1)$

Name: _____

Student ID: _____

2. (0.5 points, 0.25 points each) For each of the following Fourier transform $(X(j\omega))$, use Fourier transform properties to determine the corresponding time-domain signal $(x(t))$

2.1. $X(j\omega) = u(\omega) - u(\omega - 2)$

2.2. $X(j\omega) = \cos(2\omega) \sin(\frac{\omega}{2})$

Name: _____

Student ID: _____

OPTIONAL, FOR PRACTICE ONLY

3. Proof

$$\mathcal{F}\{e^{-|t|}\} = \frac{2}{\omega^2 + 1}$$

4. Using the outcome obtained in Problem 3., Find the Fourier Transform of the given equation.

4.1. $\frac{d}{dt}(e^{-|t|})$

4.2. $\exp(j3t - |2t + 2|)$

4.3. $\frac{1}{2\pi t^2 + 1}$

Name: _____

Student ID: _____

5. Proof

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$$

6. Determine a signal $x(t)$ which has its Fourier transform:

$$X(j\omega) = \delta(\omega) - \delta(\omega - a_0\pi) - \delta(\omega - a_1)$$

And the impulse response in time-domain is $h(t) = u(t) - u(t - 2b_0)$ 6.1. Show how to find $x(t)$ and determine whether it is periodic?6.2. Show how to find $y(t) = x(t) * h(t)$ and determine whether it is periodic?

Verify your results by sketching $x(t)$, $h(t)$, $y(t)$, $X(j\omega)$, $H(j\omega)$, and $Y(j\omega)$. Note that for the complex number, plot only its magnitude.